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Commentary on “Dynamic Incentives in Sales Force Compensation”

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Abstract. We show that the Hamilton-Jacobi-Bellman (HJB) equation for the salesperson’s problem is not correct because of the effect of a positive discount factor for the infinite-horizon risk-sensitive control. As a result, their analyses and insights are incorrect. We present the correct HJB equation for the salesperson’s problem, which is a partial differential equation. Thus, further investigation on the correct solutions is needed to better understand the problem.

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Keywords: sales force • compensation • sales dynamics • agency theory • differential games

1. Introduction

Rubel and Prasad (2016) (hereafter RP) investigate the optimal sales force compensation with sales carryover effects using a continuous-time principal-agent model. In RP, the Hamilton-Jacobi-Bellman (HJB) equation for the salesperson’s problem is the foundation of their analyses and results; see equation 3 in Rubel and Prasad (2016, p. 679). However, we show that their HJB equation is *not correct* and not consistent with their claimed value function (see equation 2 and the subsequent paragraph in Rubel and Prasad 2016, p. 679).

Note that the salesperson’s problem is indeed an *infinite-horizon* risk-sensitive control problem with discounting, whereas the references cited by RP (e.g., Jacobson 1973, Whittle 1986, Bensoussan and Elliott 1995, Başar 1999) only deal with *finite-horizon* risk-sensitive control problems without discounting. Because of the unique feature of the risk-sensitive control, the formulations for these two problems are structurally different.

2. Correct HJB Equation

In this section, we provide the correct HJB equation for the salesperson’s problem following Menaldi and Robin (2005). Define the salesperson’s value function

as follows:

$$J(\theta, x) = \max_{v(\cdot)} \mathbb{E} \left[-\exp \left(-\theta \int_0^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right], \quad (1)$$

and define its certainty equivalence as follows:

$$V(\theta, x) = -1/\theta \log(-J(\theta, x)) \\ = \max_{v(\cdot)} \left\{ -\frac{1}{\theta} \log \mathbb{E} \left[\exp \left(-\theta \int_0^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right] \right\}.$$

Note that the salesperson’s value function $J(\theta, x)$ is in fact a function of both x and the risk-aversion parameter θ . To see why the risk-aversion parameter θ is needed for the salesperson’s value function J , consider any infinitesimally small time $\tau > 0$. Based on the dynamic programming maximum principle, we have the following

equation:

$$\begin{aligned}
 J(\theta, x) &= \max_{v(\cdot)} \mathbb{E} \left[-\exp \left(-\theta \int_0^\tau e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right. \right. \\
 &\quad \left. \left. - \theta \int_\tau^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right] \\
 &= \max_{v(\cdot)} \mathbb{E} \left[-\exp \left(-\theta \int_0^\tau e^{-rt} \{S(x(t)) \right. \right. \\
 &\quad \left. \left. - v(t)^2/2\} dt \right) J(\theta e^{-r\tau}, x_\tau) \right]
 \end{aligned}$$

(see the details in the proof of Claim 1 in the appendix). This implies that the salesperson indeed has different risk-aversion parameters at time 0 and time τ because of the CARA utility. In other words, the *effective* risk-aversion parameter changes over time under CARA utility with the positive discount factor r . Therefore, the HJB equation must consider the impact of the dynamics of the effective risk-aversion parameter. The following claim provides the correct HJB equations of the salesperson's problem associated with $J(\theta, x)$ and $V(\theta, x)$, respectively.

Claim 1. Given $r > 0$, the HJB equation of the salesperson's problem associated with $J(\theta, x)$ is

$$-r\theta J_\theta + \max_v \left\{ -\theta \left(S - \frac{v^2}{2} \right) J + (v - \delta x) J_x + \frac{\sigma J_{xx}}{2} \right\} = 0, \quad (2)$$

where J_θ denotes the partial derivative with respect to the first argument and $J_x = \partial J / \partial x$, $J_{xx} = \partial^2 J / \partial x^2$. Furthermore, the HJB equation associated with $V(\theta, x)$ is

$$r(V + \theta V_\theta) = \max_v \left\{ S - \frac{v^2}{2} + (v - \delta x) V_x + \frac{\sigma V_{xx}}{2} - \frac{\sigma \theta V_x^2}{2} \right\}, \quad (3)$$

where V_θ denotes the partial derivative with respect to the first argument and $V_x = \partial V / \partial x$, $V_{xx} = \partial^2 V / \partial x^2$.

Claim 1 indicates two major issues with the salesperson's HJB equation of RP. First, the additional term J_θ or V_θ is missing in RP. As mentioned earlier, the presence of these two terms is because of the fact that the effective risk-aversion parameter changes over time under constant absolute risk aversion (CARA) utility with discounting. Unfortunately, RP ignores such dependency, leading to the incorrect HJB equation, which is only an ordinary differential equation (ODE). By sharp

contrast, one needs to solve a partial differential equation (PDE) under the correct HJB equation in Equation (2) or (3). In general, a solution to an ODE is not a solution to a PDE. As a result, the PDE is technically challenging to solve, and their analyses no longer apply.

Second, RP defines the salesperson's value function as $V(x) = J(x) = \max_{v(\cdot)} \mathbb{E}[-\exp(-\theta \int_0^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt)]$, which is not consistent with their HJB equation. Notice from Equation (3) that the term θV_x^2 only appears in the HJB equation associated with the certainty equivalence. Therefore, to be consistent, a log transformation should be included to define the salesperson's value function in their setting: that is, $V = -1/\theta \log(-J)$.

Because the HJB equation of the salesperson's problem is the foundation of the analyses in RP, Claim 1 implies that the results and insights on the equilibrium effort strategy and the optimal compensation in RP are incorrect for $r > 0$. For instance, one can verify that their solution (e.g., $v^* = (\Pi_x - \delta x)/(\theta\sigma - 1)$ for the case $1 - \theta\sigma < 0$) does not satisfy Equation (3). To obtain the optimal effort strategy and the optimal compensation plan, one shall solve the PDE in Equation (2) or (3). Unfortunately, the correct HJB equation is a second-order nonlinear PDE, and solving such an HJB equation is technically challenging. We shall leave it for future investigation. Nevertheless, RP should also be credited for introducing the continuous-time risk-sensitive control model to the salesforce compensation problem.

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Appendix. Proof of Claim 1

Proof of Claim 1. The proof in fact directly follows from section 2 in Menaldi and Robin (2005). For any $\tau > 0$, we have

$$\begin{aligned}
 J(\theta, x) &= \max_v \mathbb{E} \left[-\exp \left(-\theta \int_0^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right] \\
 &= \max_v \mathbb{E} \left[-\exp \left(-\theta \int_0^\tau e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right. \right. \\
 &\quad \left. \left. - \theta \int_\tau^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right] \\
 &= \max_v \mathbb{E} \left\{ \left[\exp \left(-\theta \int_0^\tau e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right] \right. \\
 &\quad \left. \times \mathbb{E}_{x_\tau} \left[-\exp \left(-\theta \int_\tau^\infty e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) \right] \right\} \\
 &= \max_v \mathbb{E} \left[\exp \left(-\theta \int_0^\tau e^{-rt} \{S(x(t)) - v(t)^2/2\} dt \right) J(\theta e^{-r\tau}, x_\tau) \right].
 \end{aligned}$$

Applying Ito's formula to $J(\theta e^{-r\tau}, x_\tau)$, we obtain

$$\begin{aligned} J(\theta, x) &= \max_v [1 - \theta(S - v^2/2)dt + o(dt)] \\ &\times \left[J + \left(-r\theta J_\theta + J_x(v - \delta x) + \frac{\sigma J_{xx}}{2} \right) dt + o(dt) \right] \\ &= \max_v \left\{ J(\theta, x) + \left(-\theta J(S - v^2/2) - r\theta J_\theta \right. \right. \\ &\quad \left. \left. + J_x(v - \delta x) + \frac{\sigma J_{xx}}{2} \right) dt + o(dt) \right\}, \end{aligned}$$

and by letting $\tau \rightarrow 0$, we obtain the HJB equation associated with $J(\theta, x)$ as follows (see equation 2.2 in Menaldi and Robin 2005):

$$-r\theta J_\theta + \max_v \left\{ -\theta \left(S - \frac{v^2}{2} \right) J + (v - \delta x) J_x + \frac{\sigma J_{xx}}{2} \right\} = 0,$$

where J_θ denotes the first-order partial derivative J with respect to the first argument and J_x, J_{xx} denote the first- and second-order partial derivatives of J with respect to x , respectively. Notice that the term J_θ appears because the effective risk-aversion parameter changes from θ to $\theta e^{-r\tau}$ as time changes when we apply the dynamic programming maximum principle.

We next derive the HJB equation associated with the certainty equivalence $V(\theta, x) = -\frac{1}{\theta} \log(-J(\theta, x))$. Notice that $J(\theta, x)$

$= -\exp(-\theta V(\theta, x))$; then, we have

$$\begin{aligned} J_\theta &= -V - \theta V_\theta J, \\ J_x &= -V_x \theta J, \\ J_{xx} &= \theta^2 V_{xx}^2 J - \theta V_{xx} J. \end{aligned}$$

Substituting into the HJB equation associated with J yields

$$\begin{aligned} r\theta J(V + \theta V_\theta) + \max_v \left\{ -\left(S - \frac{v^2}{2} \right) \theta J - (v - \delta x) \theta J V_x \right. \\ \left. + \frac{-\sigma \theta J(V_{xx} - \theta V_x^2)}{2} \right\} = 0. \end{aligned}$$

Cancelling out $-\theta J$ and rearranging terms obtains Equation (3):

$$r(V + \theta V_\theta) = \max_v \left\{ S - \frac{v^2}{2} + (v - \delta x) V_x + \frac{\sigma(V_{xx} - \theta V_x^2)}{2} \right\}. \quad \square$$

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